Universidad del Desarrollo
Ingeniería Civil Industrial
Business Intelligence
IIB423T-1

C1 SUBMISSION: FROM THE W1 PROJECT TO DEFENSIBLE EVIDENCE

HOW MUCH DOES THE START DECIDE A FORMULA 1 RACE? (2014–2024)

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EXECUTIVE SUMMARY

We develop Candidate B from W1: how much the starting position determines the result of a Formula 1 race in 2014–2024, whether this varies by circuit, and how much of it is really the car. The proposed user is a team strategist who, because of parc fermé, must decide how far to lean a single car setup toward one-lap (qualifying) pace or race pace at each circuit.

We worked in three passes over the public Kaggle/Ergast dataset: an EDA of the raw data (12 CSV files, 1950–2024) to decide the period and identify quality issues; a cleaning notebook that builds one validated analytic table (4,606 driver-race rows, 42 columns, 227 races); and an EDA driven by the KPI. The main KPI is the IDC (Grid Determinism Index): for each circuit, the median of the per-race Spearman correlation between grid and final position among finishers, with a 95 % bootstrap confidence interval.

Main findings: (1) the overall IDC is 0.81 (95 % CI 0.79–0.83): among finishers, the finishing order closely follows the grid; (2) once the team's championship rank before the race is controlled for, the grid–result correlation falls from 0.77 to 0.50, so a large part of the relationship is car performance; (3) circuit IDCs range from 0.68 (Monza) to 0.91 (Montmeló) but their intervals overlap, a Kruskal-Wallis test is not significant ($p = 0.29$) and only Monaco is clearly above the average; (4) the probability of scoring points is 90 % from P1–P3 and 33 % from P11–P15, and 40 % of all position changes happen on lap 1.

Implication for the proposed decision: track position is valuable at every circuit and Monaco is the one clear "qualifying circuit", but the data cannot rank the other circuits, and they observe positions and lap times, not setups. The line continues toward a pre-race classification task (top-10 finish) and a per-circuit dashboard. Supporting files: `analysis/01-03` notebooks (executed), `data/processed/` and `README.md`.

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1 ADAPTING THE CONTEXT

The C1 brief is set in an educational health-data observatory in Chile. On September 24, 2026, during the W1 guided class, the instructor authorized our Formula 1 dataset and checked the adaptation of the criteria to our domain. C1 continues the line selected in W1, so we keep the structure and requirements of the brief and change only the domain (Table 1).

Table 1. Equivalences between the brief and our project

Element of the brief	Equivalent in our project
A bounded concern, population, place and period	A competitive-performance problem in the Formula 1 World Championship: drivers and teams in 227 Grands Prix at 32 circuits, seasons 2014–2024
MINSAL priority or plan	FIA Sporting and Technical Regulations (with their year) and official F1/FIA objectives. There is no health-policy match; we state this boundary instead of inventing one
Policy objective vs numerical target vs indicator	The 2022 technical regulations state an objective (closer racing, with more overtaking as a secondary benefit) and no numerical target in the sources we consulted. Our indicator (per-race ρ, IDC) can test whether the start became less decisive, not whether racing became "closer"
Proposed user and decision	A team strategist (our proposal): how far to lean the single parc-fermé setup toward qualifying or race pace
Literature and technical evidence	Academic papers on F1 performance, FIA documents and the dataset documentation

Source: prepared by the authors.

2 THE PROBLEM AND ITS EVOLUTION FROM W1

2.1 THE PROBLEM

Table 2. The selected problem

Component	Content
Problem	Qualifying sets the starting grid. If the finishing order largely reproduces the grid, one-lap performance on Saturday decides much of Sunday's result. We ask how strong that relationship is, whether it varies by circuit, and how much of it is car performance
Population and unit	Drivers in the 227 championship Grands Prix of 2014–2024 (Belgium 2021 excluded, see chapter 4.3); unit of observation = one driver in one race
Place and period	World Championship calendar: 32 circuits, 19 of them with at least 5 races in the period; seasons 2014–2024 (hybrid-engine era)
Proposed user and use	A team strategist (our proposal) deciding, for each circuit, how far to lean the single setup toward qualifying pace or race pace
Main question	How strongly does the starting position determine the final position among finishers in 2014–2024, does it vary by circuit, and how much remains after controlling for the team's strength?

Source: prepared by the authors.

2.2 DOCUMENTARY SUPPORT

- **Qualifying predicts the result.** Using 2010–2023 data, Weissbock and Mills (2025, preprint) find that qualifying position is the best single predictor of the final position among the weekend sessions (Spearman ≈ 0.76). Our pooled value among finishers, 0.77, is consistent with it.
- **The car matters more than the driver.** Bell et al. (2016) show with multilevel models that the team effect exceeds the driver effect and has grown over time. This is why we control for team strength before interpreting the grid effect.
- **One setup for both sessions.** Under parc fermé, the car that qualifies is the car that races (FIA, 2023; FIA, 2024), so the strategist does not choose two setups but how far to lean a single one.
- **A regulatory objective, not a result.** The 2022 technical regulations had closer racing as their guiding principle, with more overtaking as a secondary benefit (Formula 1, 2021). It is an objective; chapter 5.4 tests it against the data.

2.3 FROM W1 TO C1

In W1 we compared two lines: A, pit-stop efficiency, and B, qualifying versus the result. We selected **B** because its measurement matches the question directly (grid and final position are observed for every driver), while in A the dataset records the total time in the pit lane rather than the crew's stationary time: the circuit explained about 38 % of the variation and the team only about 2 % after normalising by race. Apart from the authorization of September 24, no further instructor feedback on W1 was available before this submission, so the W1 questions remained open; Table 3 records our provisional decision for each one.

Table 3. W1 open questions and the decision taken in C1

W1 question or proposal	Decision in C1 (provisional)	Evidence	Location
Are FIA regulations an adequate equivalent of MINSAL priorities?	Adaptation validated in class on Sept 24; FIA regulations kept as policy context, boundary stated	Chapter 1	Chapter 1
Qualifying → result or start → result?	Grid (real starting order) as the main measure; qualifying position as a sensitivity check	On 2021–22 sprint weekends 84 % of grids differ from qualifying; qualifying → final gives 0.84	NB02 part 8; NB03 part 9
Retirements: exclude, separate or full order?	Finishers for the IDC; retirements reported separately and in the sensitivity analysis	DNF rate 9 % (P1–P3) to 21 % (P11+); IDC with retirements 0.65	NB03 part 5, part 9
With 7–13 races per circuit, what uncertainty is acceptable?	Show a 95 % bootstrap interval per circuit and test differences; do not rank circuits	Only Monaco's interval is above the average; Kruskal-Wallis p = 0.29	NB03 part 4
Proposed join with constructor_standings (previous race)	Implemented and validated; round 1 uses the previous season's final standings by team lineage	4,382 direct matches + 220 from the previous season; 4 unmatched	NB02 part 7, part 11

Source: prepared by the authors.

Other changes from W1: the KPI was formally defined (IDC plus support and control KPIs), a control for car strength was added, the cleaning became a separate and validated pipeline, the EDA was extended with distributions, densities and correlations, and the Five Cs were updated to the lenses of the C1 brief.

3 SEVEN-STEP MAP

Table 4. The seven steps from Class 03 applied to the project

Step	Our project	Location
1. Decision	Proposed user: a team strategist. Decision: how far to lean the single parc-fermé setup toward one-lap pace or race pace at each circuit. Alternatives: a qualifying-oriented, a race-oriented or a balanced setup	Chapter 2.1
2. Success criteria	Desired outcome: a better expected finishing position (more points) given the circuit. Measurable now: how much the starting position is worth (IDC, P(top 10 | grid)). Not measurable: the setup itself and its effect	Chapter 5.3, 5.5
3. Business questions	Main: how strongly does the grid determine the final position? Supporting: does it vary by circuit? How much remains after controlling for car strength? Where are positions won and lost (lap 1 vs the rest)?	Chapter 2.1
4. Business process	Qualifying (Q1–Q3) → penalties or sprint → grid → start and lap 1 → race (pit stops, incidents) → classification or retirement. Outside the data: setup choices, weather, safety cars, tyres and the reasons for penalties	NB03, introduction
5. Entities, events, states	Entities: driver, team (lineage), race, circuit. Events: qualifying, start, lap, pit stop, finish or retirement. States: grid slot, position after lap 1, final position, finished or classified	NB02 parts 2–8
6. Data representation	Analytic table: one row = one driver in one race (4,606 rows; key raceId + driverId). Race table: one row = one race (227 rows) for the IDC. Joins documented in Table 8	NB02 part 11; NB03 part 1
7. Measurement	ρ_race = Spearman(grid, final position) among finishers with a grid slot (≥ 8 per race); IDC = median of ρ_race over a circuit's races (circuits with ≥ 5 races) with 95 % bootstrap CI; P(top 10 | block) = top-10 finishes ÷ starters in the block (Wilson CI); pole conversion = wins from P1 ÷ races; partial ρ controls for the team's rank before the race. Exclusions: Belgium 2021; pit-lane starts are left out of rank statistics	NB03 part 1, parts 4–7

Source: prepared by the authors.

4 DATA, PROFILING AND PREPARATION

4.1 SOURCE AND WORKFLOW

Source: *Formula 1 World Championship (1950–2024)*, published on Kaggle by rohanrao with data from the Ergast API, downloaded on September 10, 2026. It has 14 CSV files linked by identifiers; we use 12 (`seasons.csv` holds only URLs and `constructor_results.csv` repeats information in `results`). Missing values are encoded as `\N`. The original files are kept unchanged in `data/raw/`; every output goes to `data/processed/`.

The work was organised in three passes, each with its own question, data and output (Table 5). Each notebook reads the output of the previous one, so the whole analysis runs from the raw files.

Table 5. Three passes over the data

Notebook	Question	Data	Output
01 · EDA of the raw dataset	What do we have?	12 CSV files, 1950–2024	Justification of the period and list of quality issues (Five Cs)
02 · Cleaning	What can we trust?	Seasons 2014–2024	Analytic table (4,606 × 42), data dictionary, cleaning log, join register
03 · KPI-driven EDA	What does it tell us?	The analytic table	KPIs with uncertainty, sensitivity checks and 5 KPI tables for the dashboard

Source: prepared by the authors.

4.2 PROFILING THE RAW DATA (NOTEBOOK 01)

We first defined the unit of observation and the key of every table and checked that the key is unique, because the key defines what a duplicate is (Table 6). All keys are unique except in `results`, where 91 (race, driver) pairs repeat; all of them are from 1950–1978, when a driver could share a car or be entered in two cars, and none falls in our period. Fourteen relationships between tables were tested and no orphan identifier was found.

Table 6. Inventory of the raw tables

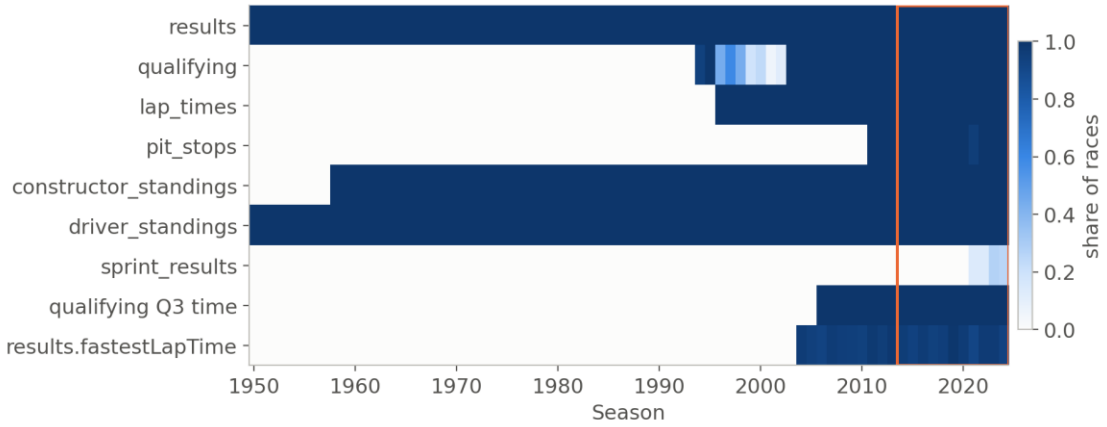
Table	One row =	Key	Rows	Dupl. keys	Years
results	one driver in one race	raceId + driverId	26,759	91	1950–2024
qualifying	one driver's qualifying	raceId + driverId	10,494	0	1994–2024
lap_times	one lap of one driver	raceId + driverId + lap	589,081	0	1996–2024
pit_stops	one pit stop	raceId + driverId + stop	11,371	0	2011–2024
sprint_results	one driver in a sprint	raceId + driverId	360	0	2021–2024
constructor_standings	team championship after a race	raceId + constructorId	13,391	0	1958–2024
driver_standings	driver championship after a race	raceId + driverId	34,863	0	1950–2024

Table	One row =	Key	Rows	Dupl. keys	Years
races, drivers, constructors, circuits, status	one entity (lookups)	own id	1,125 / 861 / 212 / 77 / 139	0	—

Source: prepared by the authors from the dataset (notebook 01, parts 1–2).

Figure 1.

Share of each season's races with data, by table (orange frame: 2014–2024)

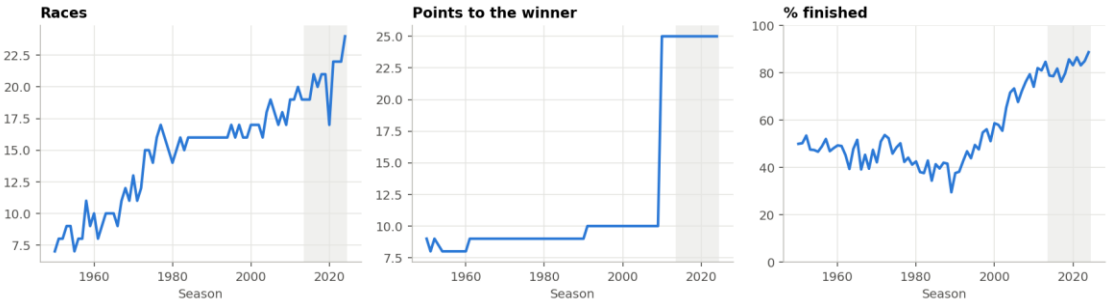


Source: prepared by the authors from the Formula 1 World Championship dataset (Rohanrao, n.d.).

Figure 1 decided the period. `results` and the standings cover the whole history, but the tables that describe how a result was built appear late: qualifying is complete from 2003, the three-session format with a Q3 time from 2006, lap-by-lap data from 1996 and pit stops from 2011. Coverage is not enough: the numbers must also mean the same thing over time (Figure 2). The winner received 8–10 points for decades and 25 since 2010, so our KPIs use positions, not points; and before the 1990s fewer than half of the starters finished, against about 80–90 % in recent seasons. We therefore kept 2014–2024: one engine formula, one points scale, knockout qualifying and every table we need complete.

Figure 2.

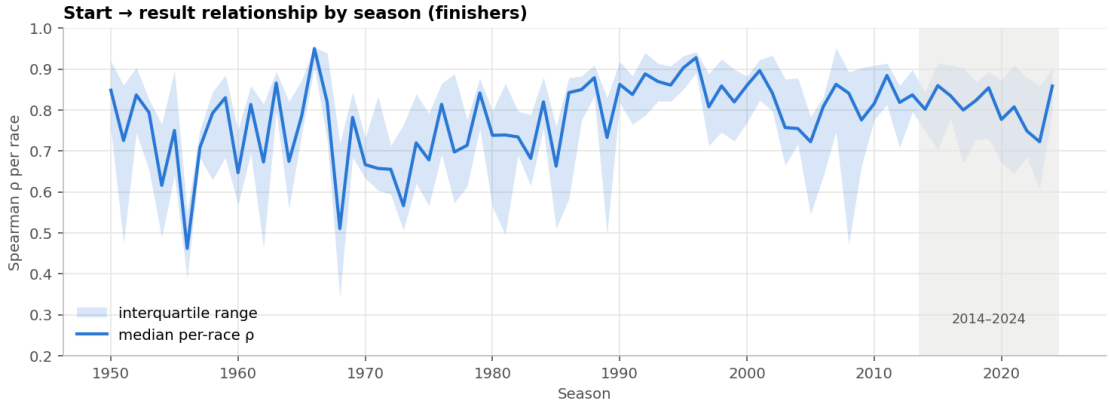
Races per season, points to the winner and share of starters who finished, 1950–2024



Source: prepared by the authors from the Formula 1 World Championship dataset (Rohanrao, n.d.).

Figure 3.

Median per-race Spearman ρ(grid, final position) by season, finishers, 1950–2024 (band: interquartile range)



Source: prepared by the authors from the Formula 1 World Championship dataset (Rohanrao, n.d.).

Before narrowing the period we looked at the question across the whole history (Figure 3). The relationship is strong in every era and has been stable at about 0.75–0.9 since the late 1980s. The Spearman matrix of `results` for 2004–2024 (notebook 01, part 7) added a warning: the final position correlates 0.77 with the grid and 0.73 with the fastest-lap rank, and those two correlate 0.64 with each other. Fast cars qualify well and race well, so a control for the car was needed.

4.3 PREPARATION (NOTEBOOK 02)

Rule of work: every preparation decision comes from a finding in notebook 01 and is checked with the data. The analysed population changes as follows: 26,759 raw results → 4,626 in 2014–2024 → 4,606 after excluding Belgium 2021 (227 races) → 3,751 finishers with a grid slot, the population of the IDC. Table 7 lists the consequential decisions.

Table 7. Preparation decisions

Decision	Reason	Alternative rejected	Check / effect
Master filter on races.raceId, 2014–2024	All tables describe the same races	2011–2024	26,759 → 4,626 results
Exclude races with < 5 laps (Belgium 2021)	1 lap behind the safety car: the order copies the grid (ρ = 1 by construction)	Keep it	−20 rows; lap_times shows 1 lap
grid = 0 → pit-lane flag, grid = NaN	A pit-lane start is not a position smaller than 1	Treat as 0 or delete	77 rows flagged and kept
fastest_lap_rank = 0 → NaN	Code for "no timed lap"	Keep the 0	217 rows
Separate finished and classified	A driver can be classified without finishing (≥ 90 % of the distance)	One flag	81 classified without finishing
Qualifying time → seconds → % gap to fastest	Comparable across circuits of different length	Absolute seconds	105 gaps > 7 % flagged, not deleted
Race pace = median green lap	Measures pace, not incidents: no lap 1, no pit in/out laps, no laps > 107 % of the race median	Mean of all laps	85.8 % of laps kept; ≥ 50 % of laps required

Decision	Reason	Alternative rejected	Check / effect
Team rank after the previous round	The standings after the same race already contain the result (leakage)	Same-race standings	99.9 % available
Round 1: previous season's final standings, by lineage	No previous round within the season	Leave empty	220 rows filled; 4 remain empty
20 constructor ids → 12 lineages	Force India → Racing Point → Aston Martin is one team	Original names	Lineage map (from W1)
No imputation	Every remaining gap has a structural reason	Impute means	Missing-value profile (Table 10)

Source: prepared by the authors.

Table 8. Join register

Join	Key	Cardinality (validated)	Matched	Unmatched and why	Rows
results → races, status	raceId; statusId	many-to-one	all	none	unchanged
results → qualifying	raceId + driverId	one-to-one	4,590 of 4,606	16: no qualifying time set	4,606 → 4,606
lap_times → pit in/out laps	raceId + driverId + lap	many-to-one	flag only	laps without a stop	248,124 laps unchanged
→ position after lap 1	raceId + driverId	one-to-one	97.0 %	did not complete lap 1	4,606 → 4,606
→ race pace	raceId + driverId	one-to-one	88.7 % (99.9 % of finishers)	retired before half distance	4,606 → 4,606
→ constructor standings, previous round	year + round − 1 + constructorId	many-to-one	4,382 + 220 (previous season)	4: team without previous standings	4,606 → 4,606
→ drivers, circuits	driverId; circuitId	many-to-one	all	none	4,606 → 4,606

Source: prepared by the authors (notebook 02, part 11). Every join uses pandas validate= and a row-count assertion.

Consistency of "grid". On the sprint weekends of 2021–2022 the Saturday sprint set the Grand Prix grid: there, 84.1 % of grids differ from the qualifying order (113 results), against 28.9 % on standard weekends (penalties) and 27.2 % on the sprint weekends of 2023–2024. The grid is therefore used as the real starting order, the qualifying position only as a sensitivity check, and sprint weekends are flagged.

Table 9. Validation checks on the analytic table (all passed)

Check	Result
Key (raceId, driverId) unique · one winner per race · final position 1..n in every race	passed
Grid between 1 and 26 · grid empty only for pit-lane starts · positions gained = grid − final position	passed
Qualifying and race-pace gaps ≥ 0 · Belgium 2021 excluded	passed

Source: prepared by the authors.

Table 10. Missing values in the analytic table and their structural reason

Column	% missing	Reason
position	15.7	Not classified (retired before 90 % of the distance)
race_pace_gap_pct	11.3	Retired before completing half of the laps
gained_lap1 / pos_lap1	4.5 / 3.0	Did not complete lap 1 (or started from the pit lane)
fastest_lap_rank	4.3	No timed lap
grid, positions_gained	1.7	Pit-lane start (flagged in pitlane_start)
quali_gap_pct	1.5	No qualifying time
constructor_rank_prev	0.1	Team with no previous standings (Haas, first race; Manor, round 2 of 2015)

Source: prepared by the authors.

Recorded zeros and missing records were kept apart: `grid = 0` and `fastest_lap_rank = 0` are codes, not values, and an empty `position` means "not classified", not an error. Because every gap has a meaning, nothing was imputed.

4.4 FIVE CS

Table 11. The five lenses of the C1 brief applied to the raw and prepared data

C	Finding in the raw data (NB01)	Handling in the prepared table (NB02)
Clean	\N as missing; grid = 0 is a code; 91 shared-car duplicates (1950–1978)	Belgium 2021 excluded (−20 rows); pit-lane starts flagged (77); rank 0 → NaN (217); 105 extreme qualifying gaps flagged; 9/9 checks pass
Consistent	Meaning of grid = 0, points scale (8–10 → 25), qualifying format (Q3 from 2006) and reliability changed over time	Period 2014–2024 with one engine formula and points scale; positions instead of points; grid = real starting order, sprint weekends flagged
Conformed	0 orphan ids in 14 relationships; the same team appears under several names and ids	All joins validated without lost or duplicated rows; 20 ids → 12 lineages; team strength from the previous round
Current	Data end with the 2024 season (Ergast stopped updating)	Suitable for a historical decision; cut-off stated
Comprehensive	Qualifying complete from 2003, pit stops from 2011; no weather, safety-car, tyre or setup fields	227 races, 59 drivers; team rank 99.9 %, race pace 99.9 % of finishers; unobserved factors declared as limitations

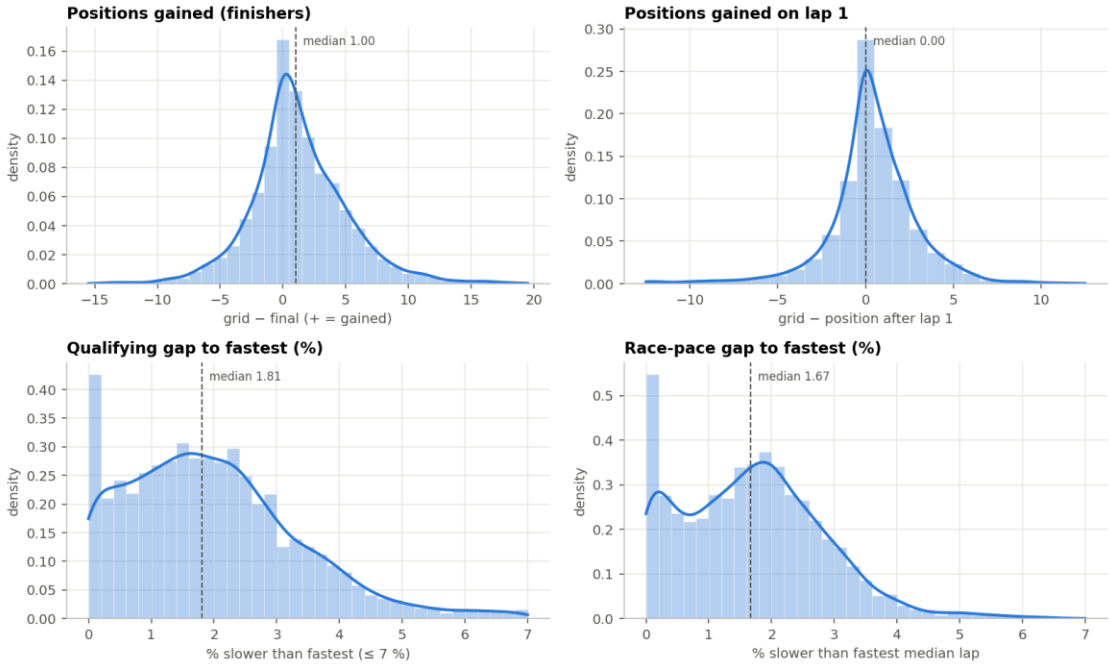
Source: prepared by the authors.

5 EDA RESULTS AND INTERPRETATION

5.1 DISTRIBUTIONS

Figure 4.

Histograms (bars) and kernel density estimates (lines) of the driver-race variables, 2014–2024



Source: prepared by the authors from the Formula 1 World Championship dataset (Rohanrao, n.d.).

Table 12. Shape of the main variables

Variable (unit)	n	Median	IQR	Skew	Reading
Positions gained, finishers (positions)	3,751	+1	4	0.37	Bounded by the grid slot; finishers gain when cars ahead retire
Positions gained on lap 1 (positions)	3,751	0	3	−1.49	Most keep their place; the big moves are losses (kurtosis 8.0)
Qualifying gap to fastest (%)	4,434	1.81	1.85	0.80	Right-skewed; most within 1–3 %
Race-pace gap to fastest (%)	4,087	1.67	1.64	0.82	Tighter than qualifying: a median of many laps
Per-race ρ(grid, final) (race level)	227	0.811	0.192	−1.67	Long left tail, capped at 1; Shapiro-Wilk W = 0.87, $p < 0.001$

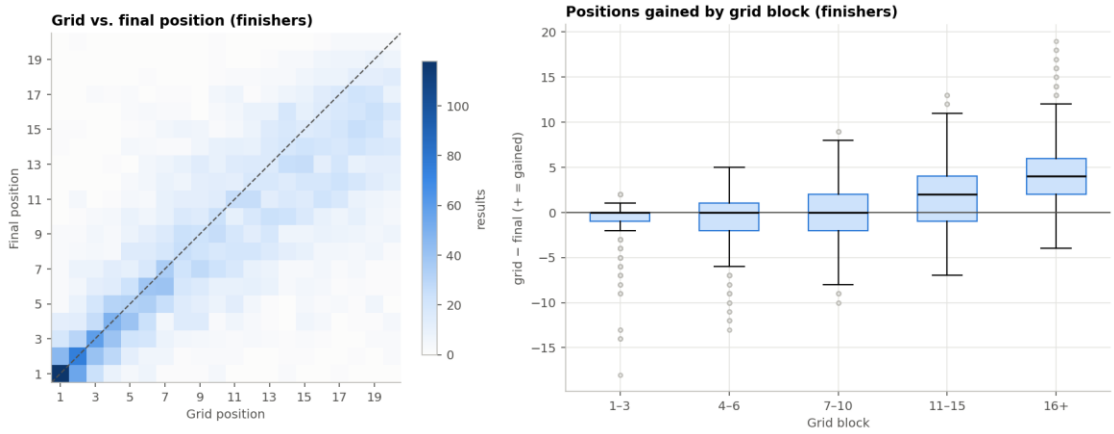
Source: prepared by the authors (notebook 03, part 2).

All the variables are skewed or heavy-tailed, and the per-race correlation, the raw material of the KPI, is clearly not normal. This decided the methods of the rest of the EDA: medians and interquartile ranges instead of means and standard deviations, bootstrap intervals, and rank-based tests (Spearman, Kruskal-Wallis, Mann-Whitney) instead of Pearson and ANOVA. Race volatility (mean absolute positions gained per finisher, median 2.8) correlates −0.79 with the per-race ρ, so it is the same KPI expressed in positions.

5.2 RELATIONSHIPS

Figure 5.

Grid vs final position among finishers (left) and positions gained by grid block (right)

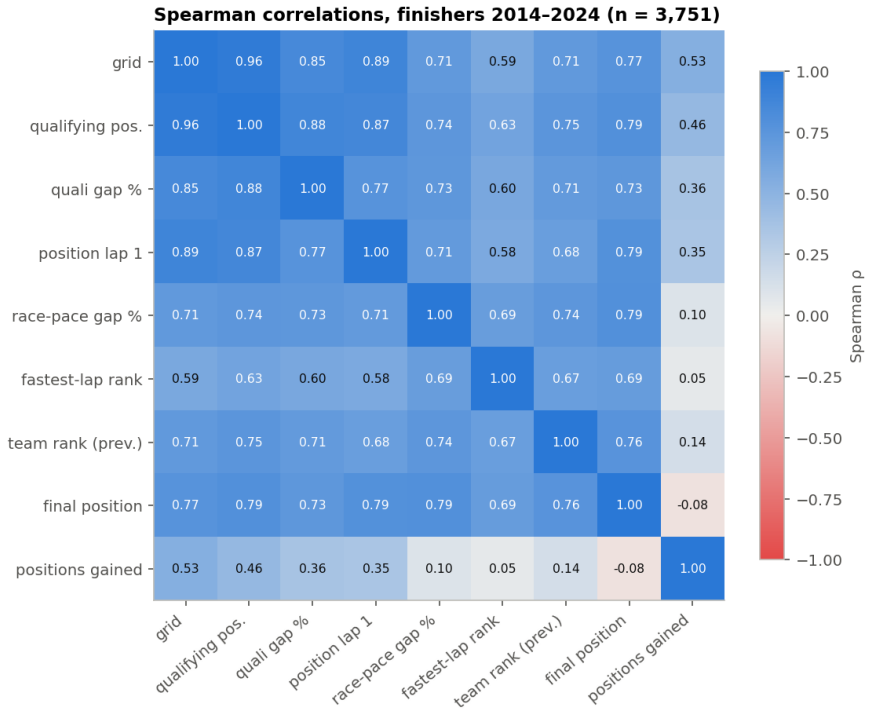


Source: prepared by the authors from the Formula 1 World Championship dataset (Rohanrao, n.d.).

The mass of results sits on the diagonal, especially at the front. The boxplot shows a floor and ceiling effect: drivers starting P1–P3 can mostly only hold or lose places (mean −0.9), while those starting 16th or lower mostly gain (median +4). Positions gained is therefore not a skill score to compare across grid blocks, and the KPIs condition on the grid.

Figure 6.

Spearman correlations among finishers, 2014–2024 (n = 3,751)



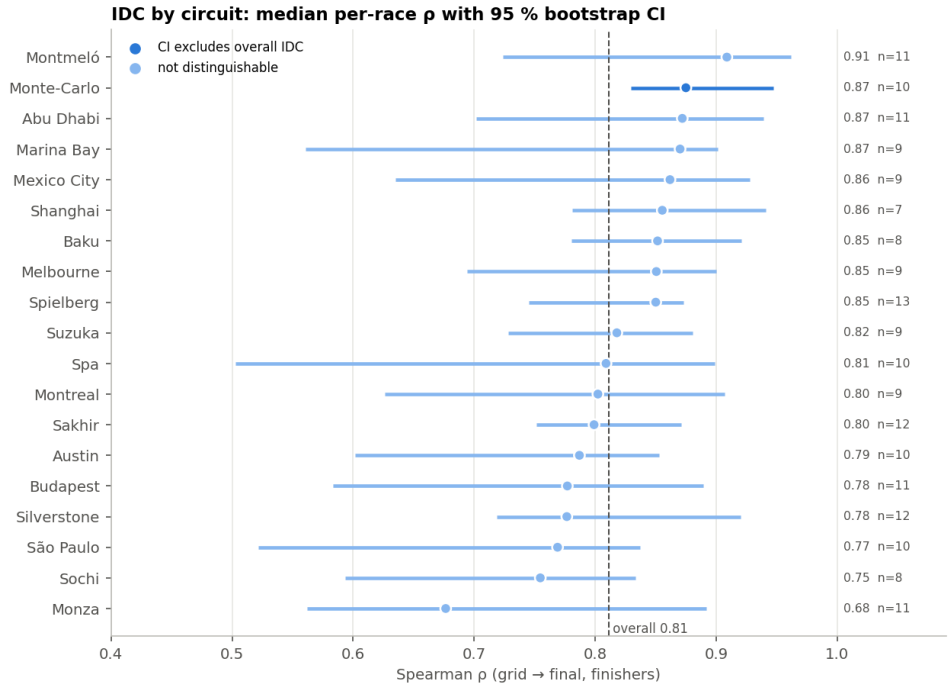
Source: prepared by the authors from the Formula 1 World Championship dataset (Rohanrao, n.d.).

The final position correlates almost equally with race pace (0.79), qualifying position (0.79), position after lap 1 (0.79), grid (0.77) and the team's rank before the race (0.76). Everything that describes a fast car moves together (for example, grid and team rank correlate 0.71). Two consequences follow: the grid effect must be separated from the car (chapter 5.6), and a future model will face multicollinearity.

5.3 MAIN KPI: IDC BY CIRCUIT

Figure 7.

IDC by circuit with 95 % bootstrap interval; dashed line: overall value (circuits with ≥ 5 races)



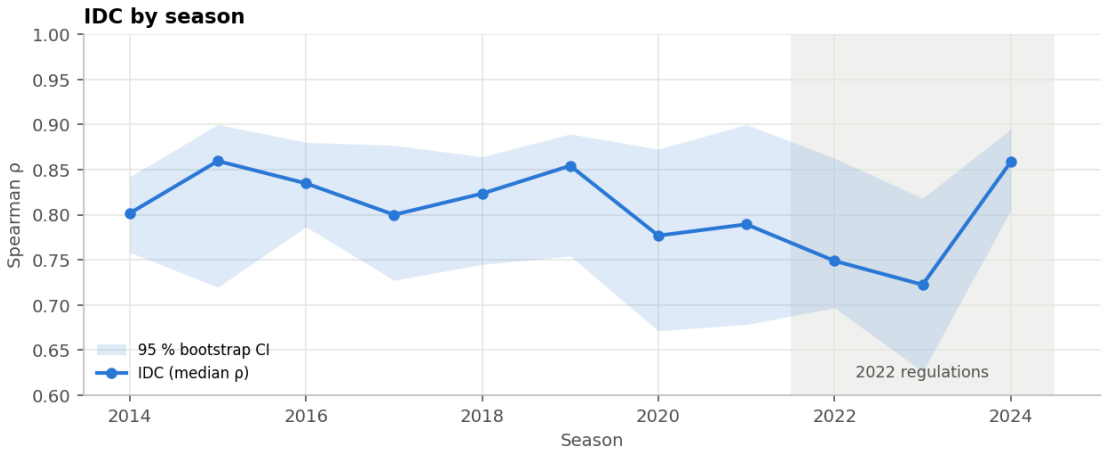
Source: prepared by the authors from the Formula 1 World Championship dataset (Rohanrao, n.d.).

The overall IDC is 0.81 (95 % CI 0.79–0.83, 227 races): in a typical race, the finishing order of the finishers closely reproduces the grid. By circuit, the median ranges from 0.68 (Monza) to 0.91 (Montmeló), but the intervals are wide (0.12 to 0.40) and overlap for almost every pair. Only Monaco (Monte-Carlo) has an interval entirely above the overall value (0.83–0.95). The Kruskal-Wallis test does not find a significant difference between circuits ($H = 20.8$, $p = 0.29$; $\varepsilon^2 \approx 0.11$). With 7–13 races per circuit we cannot rank circuits; this is an appropriately inconclusive result, not a failure of the KPI.

5.4 THE IDC OVER TIME AND THE 2022 REGULATIONS

Figure 8.

IDC by season with 95 % bootstrap interval; grey: seasons under the 2022 technical regulations



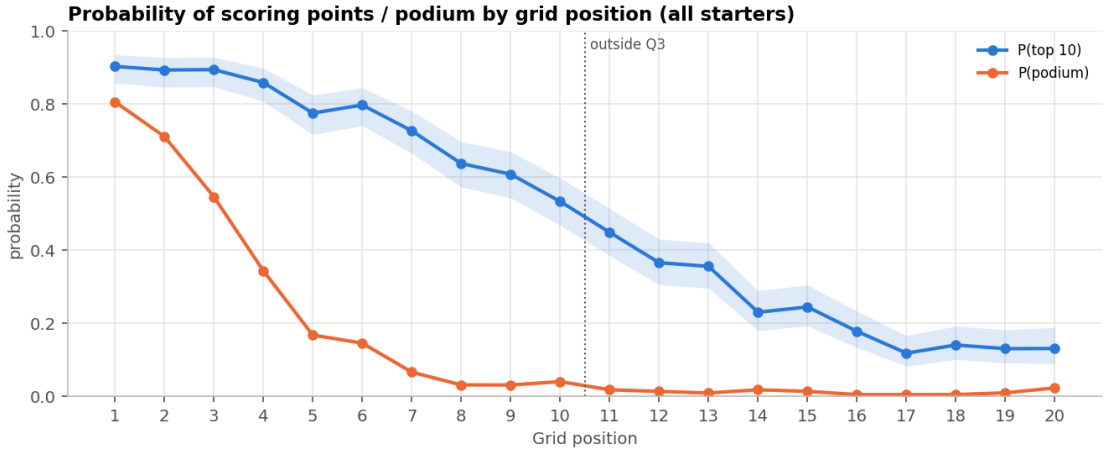
Source: prepared by the authors from the Formula 1 World Championship dataset (Rohanrao, n.d.).

The IDC dips in 2020–2023 (lowest 0.72 in 2023) and returns to 0.86 in 2024. The median per-race ρ is 0.818 in 2014–2021 (159 races) and 0.797 in 2022–2024 (68 races), a difference that is not distinguishable (Mann-Whitney p = 0.30). The regulatory objective of closer racing is therefore not reflected in a less decisive start, although this comparison does not control for team dominance or calendar composition.

5.5 SUPPORT KPIS: THE VALUE OF A GRID SLOT

Figure 9.

Probability of a top-10 finish (blue, 95 % Wilson band) and of a podium (orange) by grid position, all starters



Source: prepared by the authors from the Formula 1 World Championship dataset (Rohanrao, n.d.).

Table 13. Outcomes by grid block, all starters, 2014–2024

Grid block	Starters	P(top 10)	P(podium)	P(win)	DNF rate
1–3	679	0.897	0.688	0.290	0.090
4–6	681	0.811	0.219	0.029	0.129
7–10	905	0.627	0.042	0.008	0.167
11–15	1,128	0.329	0.014	0.002	0.211
16+	1,136	0.138	0.010	0.001	0.211
Pit lane	77	0.182	0.000	0.000	0.299

Source: prepared by the authors (notebook 03, part 5).

The probability of scoring points is about 0.90 from P1–P3 and about 0.13 from P17 back. Between P7 and P14 it falls from 73 % to 23 %, about 7 percentage points per grid slot on average: those are the slots where one qualifying position is worth the most. Starting further back also carries more retirement risk (9 % in the first three places vs 21 % from 11th). Pole conversion is 52 % (95 % Wilson CI 45–58 %); by circuit it ranges from 25 % (Baku, Sochi) to 91 % (Abu Dhabi), but with 7–13 races the intervals are too wide to rank circuits, so it is used only as a headline.

5.6 CONTROL KPI: IS IT THE START OR THE CAR?

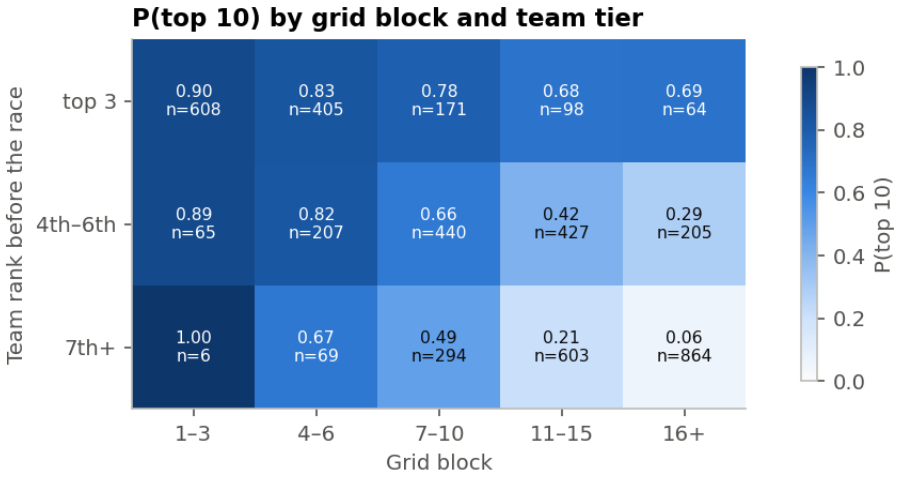
Table 14. Grid–result association with and without team strength (finishers with a team rank; CIs from a bootstrap of whole races)

Measure	Estimate	95 % CI
ρ(grid, final position)	0.769	0.749–0.789
ρ(grid, team rank before the race)	0.715	—
ρ(final position, team rank before the race)	0.758	—
Partial ρ(grid, final | team rank)	0.498	0.464–0.533
ρ(grid, final) within tier: top 3 / 4th–6th / 7th+	0.634 / 0.568 / 0.546	—

Source: prepared by the authors (notebook 03, part 7).

Figure 10.

P(top 10) by grid block and team rank before the race (n per cell shown)



Source: prepared by the authors from the Formula 1 World Championship dataset (Rohanrao, n.d.).

Once the team's pre-race rank is removed, the association falls from 0.77 to 0.50: in shared variance (ρ^2), from about 59 % to 25 %. A large part of "qualifying decides the race" is really "the car decides both". The remaining 0.50, with an interval well above zero, and the positive correlation inside every tier show that track position still matters on its own. Figure 10 adds the practical nuance: a top-3 team starting 11th–15th still scores 68 % of the time, while a team ranked 7th or lower starting 16th or further back scores 6 %; a fast car recovers from a bad grid slot, a slow one does not.

5.7 Where positions change: lap 1

Among finishers, 40.4 % of all position changes (in absolute value) happen on the first lap. The order after lap 1 already correlates 0.79 with the final order (grid → lap 1: 0.89). By circuit, the lap-1 share ranges from 29 % (Monaco) to 46 % (Sochi, Montmeló) and is only weakly related to the IDC ($\rho = -0.19$).

5.8 Robustness and outliers

Table 15. Sensitivity of the overall IDC to measurement decisions

Variant	Median ρ	Races	Reading
Main: finishers, grid → final	0.811	227	Reference
All starters including retirements	0.654	227	Retirements break the link; reported separately
Qualifying position → final (finishers)	0.836	227	Higher: grid penalties often hit fast cars that recover
Excluding sprint weekends	0.818	209	Robust
Kendall τ instead of Spearman	0.650	227	Different scale, same ordering
Only circuits with ≥ 5 races	0.821	189	Robust

Source: prepared by the authors (notebook 03, parts 9–10).

Seven races fall below the IQR lower fence ($\rho < 0.41$), among them Germany 2019 ($\rho < 0$), Turkey 2020 and São Paulo 2024, all wet races. They are real outcomes, not data errors, and removing them only moves the overall IDC from 0.811 to 0.820, so they were kept.

5.9 OBSERVATION, INTERPRETATION AND HYPOTHESIS

Table 16. What the evidence shows and what it suggests

Observed	Interpretation	Hypothesis for later
Overall IDC 0.81 (0.79–0.83)	Among finishers, the finishing order closely follows the grid	—
ρ falls from 0.77 to 0.50 with team rank	Much of the association reflects car performance	A driver-level pace measure (team-mate comparison) would reduce it further
Qualifying → final 0.84 > grid → final 0.81	Penalised fast cars recover through the field	Testable with penalty data, not in the dataset
Circuit intervals overlap; only Monaco is above	Circuit differences are not established with 7–13 races	More seasons or a model that pools circuits
Partial ρ 0.47 (2014–21) vs 0.55 (2022–24)	Exploratory: once the car is removed, the start may matter slightly more	Requires a formal test
40 % of position changes on lap 1	The start phase is decisive	Weather and safety-car data would separate incidents from racing

Source: prepared by the authors.

6 CONCLUSIONS, IMPLICATIONS AND LIMITATIONS

6.1 ANSWER TO THE QUESTION

As far as the evidence allows: in 2014–2024 the starting position strongly determines the final position among finishers (IDC 0.81), consistently across seasons; a large part of this is car performance (0.77 → 0.50 after controlling for team strength), but track position keeps an effect of its own; and differences between circuits are not established, with Monaco as the only circuit clearly above the average.

6.2 IMPLICATIONS FOR THE PROPOSED DECISION

- Qualifying performance is valuable at every circuit: between P7 and P14 each grid slot is worth about 7 points of probability of scoring, and 40 % of position changes happen on lap 1.
- Monaco is the one circuit where the evidence supports leaning clearly toward qualifying; for the rest, the data support a general rule, not a circuit ranking.
- For a slow car the grid matters more than for a fast one, which recovers from a bad slot (Figure 10).

6.3 ALTERNATIVE EXPLANATIONS AND WHAT THE ANALYSIS CANNOT ESTABLISH

The data record positions and lap times, **not setups**: the analysis measures how much the starting position is worth, which is an input to the strategist's decision, but it cannot show the effect of a given setup. All results are associations: car performance (partly controlled), reliability, penalties, weather and safety cars can each produce the same patterns, and the last two are not recorded. Team strength is measured with the constructors' championship rank, a coarse proxy that does not capture in-season upgrades or differences between team-mates.

Table 17. Limitations and their consequences

Limitation	Consequence	Mitigation
No weather, safety-car or tyre data	Chaotic races stay in the sample	Median; sensitivity without outliers (0.811 → 0.820)
Team strength = constructors' rank before the race	Coarse proxy of car performance	Still reduces ρ from 0.77 to 0.50; driver-level pace in the next stage
7–13 races per circuit	Wide intervals; no circuit ranking	Intervals shown explicitly; more seasons would narrow them
Qualifying gap mixes dry and wet sessions	Noise in that variable	Used as a support variable; the KPI uses positions
Data end in 2024	Current regulations after 2024 not covered	Cut-off stated; suitable for a historical decision

Source: prepared by the authors.

Further evidence needed: weather and track-status (safety car) data, tyre strategies, penalty records and a driver-level pace measure would allow separating the start effect from incidents and from the car more precisely.

7 PROJECT CONTINUITY: ML AND DASHBOARD

Neither a model nor a dashboard is implemented in C1; this section shows that the line has a plausible, concrete continuation grounded in the EDA.

Table 18. Proposed machine-learning task

Element	Proposal
Task and decision value	Supervised classification: will a driver finish in the top 10? It estimates, before the race, how many points a grid position is likely worth for a given team and circuit
Unit of analysis and outcome	One driver in one race; outcome top10, defined from the final position (about half of the rows by construction)
Inputs available before the race	grid, quali_pos, quali_gap_pct, pitlane_start, constructor_rank_prev, team_tier, circuit, era, sprint_weekend
Never inputs (known only after the start)	pos_lap1, gained_lap1, race_pace_gap_pct, fastest_lap_rank, finished, status, points
Baseline	"You finish where you start": P(top 10 | grid) from Table 13
To review in the next stage	Multicollinearity among pace and position variables (Figure 6); testing on seasons not used for training; missing data that would help (weather, safety car)

Source: prepared by the authors.

Table 19. Proposed dashboard

Element	Proposal
User and decision	The team strategist before a race weekend: how much to prioritise qualifying at the next circuit
Core indicators	IDC with its interval, P(top 10) by grid position, pole conversion, lap-1 share of position changes, retirement rate
Comparisons and filters	Circuit vs overall value; season or era; team tier
Data	kpi_summary_by_circuit.csv and kpi_P_top10_by_grid_block.csv (data/processed)
How the EDA shapes it	It must show the intervals, because most circuits are not distinguishable, and separate team tiers, because the value of a grid slot depends on the car

Source: prepared by the authors.

8 PROCESS RECORD

8.1 DECISIONS AND CHECKS

Table 20. Record of decisions and checks

Date / stage	Decision	Alternative	Reason and evidence	Check / result	Location
Sep 26 · scope	Keep 2014–2024	Whole history; 2011–2024	Coverage and comparability	Coverage heatmap; points; finishing rate	NB01 parts 3–4
Sep 26 · KPI	IDC: median per-race Spearman with bootstrap CI	Pooled correlation; pole conversion as main KPI	Ordinal, skewed data; 7–13 races per circuit	Shapiro-Wilk $p < 0.001$	NB03 part 2, part 4
Sep 27 · cleaning	Exclude races with < 5 laps	Keep Belgium 2021	$\rho = 1$ by construction	−20 rows	NB02 part 3
Sep 27 · cleaning	grid = 0 → pit-lane flag	Treat as 0	Not a position	77 rows flagged	NB02 part 4
Sep 27 · control	Team rank after the previous round	Same-race standings	Avoid leakage	99.9 % coverage; validated join	NB02 part 7
Sep 27 · measure	Grid as main measure	Qualifying position	Real start; sprint grids	Qualifying variant 0.84	NB02 part 8; NB03 part 9
Sep 27 · measure	Finishers for the IDC	All starters	Retirements break the link	Variant 0.65; DNF by block	NB03 part 5, part 9
Sep 27 · control	Partial ρ with bootstrap of whole races	Plain correlation	Drivers of a race are not independent	0.50 (0.46–0.53)	NB03 part 7
Sep 27 · outliers	Keep chaotic races	Delete them	Real events; median robust	IDC 0.811 → 0.820	NB03 part 10
Sep 28 · rubric	Five Cs updated to the C1 lenses; join register; seven-step table	Keep the W1 lists	C1 brief	Notebooks re-executed	NB01 part 9; NB02 part 8, part 11; NB03
Sep 28 · text	Corrected three interpretation sentences of notebook 03	Keep them	Imprecise against the outputs	Outputs unchanged	NB03 part 4, part 5, part 10

Source: prepared by the authors.

8.2 INSTRUCTOR FEEDBACK AND RESPONSE

On September 24, 2026, during the W1 guided class, the instructor authorized the use of the F1 dataset and checked that the adaptation of the criteria from the MINSAL data to our domain was correct. We accepted this guidance and keep the equivalence table (chapter 1). No further feedback on the W1 questions was available before this submission; the questions are therefore still pending, and Table 3 records our provisional decision and its evidence for each one.

8.3 INDIVIDUAL CONTRIBUTIONS

C1 was a joint project: both members worked on every stage, took the main decisions together in shared work sessions and reviewed each other's parts. Table 21 shows how the work was divided within that shared process.

Table 21. Individual contributions to C1

Member	Contribution	Evidence
Both members	Selection of the line and the KPI after W1; definition of the period 2014–2024; review of every cleaning decision; interpretation of the results; review of the report, the slides and the script; rehearsal of the defense	Chapters 2–6; decision log (Table 20)
Maximiliano Sandoval	Led the framing of the problem and its evolution from W1 (context adaptation, documentary support, seven-step map); led the EDA of the raw dataset (notebook 01: inventory, integrity, coverage and comparability) and the Five Cs; wrote chapters 1–3 and 4.1–4.2 of the report; presents slides 1–7	Notebook 01; chapters 1–3, 4.1–4.2; slides 1–7
Benjamín Toloza	Led the cleaning and the KPI-driven EDA (notebooks 02 and 03: joins, validations, IDC, support and control KPIs); ran the three notebooks from top to bottom and checked the outputs; coordinated the AI-assisted work and its verification; wrote chapters 4.3–4.4 and 5–7 of the report; presents slides 8–16	Notebooks 02 and 03; chapters 4.3–7; slides 8–16

Source: prepared by the authors.

8.4 AI USE AND VERIFICATION

Tool: Claude (Anthropic), in Cowork mode. **Purpose:** recommend data and KPIs, draft and review the notebooks, prepare the presentation and script, review the work against the C1 rubric and draft this report. All data are public; nothing confidential was uploaded. No AI will be used during the defense.

Table 22. AI use and verification

Use	Prompt excerpt (translated)	Accepted, modified or rejected	How we verified it
Data needed	"Tell me which databases or .csv files you will need to make an EDA"	Accepted: add constructor_standings, drivers, constructors, lap_times, driver_standings	Coverage checked by table and year (NB01 part 3)
KPIs	"Recommend KPIs related to this line of inquiry"	Accepted the IDC with support and control KPIs; preliminary values modified (partial ρ 0.49 → 0.50 after the round-1 fill)	All values recomputed in NB03
Notebooks	"Start with the notebook for the requested EDA… two EDAs"	Modified: three notebooks (cleaning separated from the two EDAs)	Restart & Run All in local Jupyter (Python 3.13.7); outputs identical; IDC 0.811 and pole conversion 52 % recomputed with an independent script
Interpretation review	Review of the notebook readings against the outputs	Accepted: three imprecise sentences corrected (value per slot, Montmeló, wet races)	Compared with the printed outputs
Rubric review	"Read the C1 rubric… how well have we done"	Accepted: report and slides PDF, new Five Cs, join register, seven steps, measurement boundary	Implemented in NB01 part 9, NB02 part 8 and part 11, NB03
Presentation and report	"Generate a base for the presentation and a script… a .docx with the key points"; report drafting	Accepted: the base for the presentation and draft for the report; Modified: the .pptx to make it more appealing and graphic for the audience.	Checked every number and information in the base against what is in the notebooks and report

Source: prepared by the authors.

8.5 INDIVIDUAL REFLECTION

Maximiliano Sandoval: "My main contribution in C1 was turning the W1 idea into a problem that could actually be measured and exploring the raw data before trusting it. Building the coverage heatmap was the moment the project made sense to me: I had assumed that more years meant a better analysis, and the data showed the opposite, because the qualifying and pit-stop tables do not exist for most of the history and the points system changed. What I learned most is that a strong correlation can hide a third factor: the grid and the result looked almost the same thing until we controlled for the car, and the relationship fell from 0.77 to 0.50. An uncertainty I still have is whether the constructors' championship position is a good enough measure of how fast a car really is, especially in seasons where a team improves a lot during the year."

Benjamín Toloza: "My main contribution was the cleaning and the EDA of the KPI and making sure that everything ran from start to finish on our own computer. I learned that most of the work in data analysis happens before the first chart: deciding what one row represents, checking every join so that no row is lost or duplicated, and understanding why a variable that is only known after the start, such as the position on lap 1, cannot be used to explain a decision taken before the race. It was also the first time I used bootstrap intervals, and they changed how I read the results: without them I would have ranked the circuits, but with them I saw that most differences are within the noise. I am still unsure whether those circuit differences would become clear with more seasons of data, or whether they are simply smaller than the effect of the car."

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